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TRANSFER CREDIT MATCH – Data Science Branch
Proof of Concept Documentation
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BRANCH: Data_Science
LAST UPDATED: April 2026
TEAM ROLE: Data Science / Database Pipeline
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OVERVIEW
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This branch contains the Data Science layer of the Transfer Credit Match system.
The goal of this component is to support intelligent course-to-course matching across institutions using structured academic data, knowledge unit (KU) mappings, and a PDF transcript parser.

The system helps students who transfer between Chicago-area colleges identify which courses they have already completed and which ones may satisfy requirements at their destination institution.

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FOLDER STRUCTURE
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```

```
Data_Science/
├── readme.txt                ← You are here
├── TranscriptPDF_reader/
│   └── pdf_reader.py        ← PDF transcript parser script
├── csv_exports/
│   ├── institutions.csv     ← Participating institutions
│   ├── programs.csv        ← Academic programs per institution
│   ├── courses.csv         ← Course catalog across institutions
│   ├── knowledge_units.csv  ← KU taxonomy (CAE-aligned)
│   ├── course_ku.csv       ← Course ↔ KU mapping (many-to-many)
│   ├── students.csv        ← Student enrollment records
│   ├── transfer_requests.csv ← Student-submitted transfer requests
│   └── course_match.csv     ← Matched course pairs (with status)
```

— match_history.csv	← Historical match audit log
— admins.csv	← Admin user records
— directors.csv	← Program director records
— users.csv	← All system users

INSTITUTIONS (Proof of Concept)

The system currently seeds four real Chicago-area institutions:

ID	Institution	Location
1	Roosevelt University	Chicago, IL
2	National Louis University	Chicago, IL
3	University of Chicago	Chicago, IL
4	Harold Washington College	Chicago, IL

ACADEMIC PROGRAMS

Each institution offers one or more programs that students enroll in:

Program ID	Institution	Program Name
1	Roosevelt University	Computer Science
2	Roosevelt University	Cybersecurity
3	Harold Washington College	Information Technology
4	Harold Washington College	Software Development
5	National Louis University	Computer Forensics
6	National Louis University	Computer Science
7	University of Chicago	Cloud Computing
8	University of Chicago	Information Technology
9	Roosevelt University	Cyber Security & Info

Assurance

COURSE CATALOG (Sample – 25 Courses Seeded)

Courses are tied to institutions and programs. Below is a sample:

Course ID Credits	Code	Course Name	
1	CS101	Data Structures and Algorithms	4
2	CYB201	Network Security	3
3	IT101	IT Fundamentals	3
4	SD102	Web Development	3
5	CSIA 150	Computer Science I	4
6	CSIA 217	Intro to Prob. & Stats	3
7	CSIA 236	Python Script Programming	3
8	CSIA 246	Data Communications	3
9	CSIA 250	Computer Science II	4
10	CSIA 255	Open Source Communities	3
11	CSIA 261	Computer Organization	3
12	CSIA 301	Computer Networking	3
13	CSIA 317	Operating Systems	3
14	CSIA 318	UNIX and System Administration	3
15	CSIA 319	Cyber Ops	3
16	CSIA 327	Software Engineering	3
17	CSIA 333	Database Systems	3
18	CSIA 335	Ethical Hack & Countermeasures	3
19	CSIA 336	Practical Computing with Data/Python	3
20	CSIA 352	Network Design	3
...	...	(+ 5 more courses seeded)	
...			

KNOWLEDGE UNITS (KU TAXONOMY – CAE-Aligned)

Knowledge Units represent standardized academic competency areas. Each course is mapped to one or more KUs to enable semantic matching across institutions.

KU ID	Knowledge Unit Name	Description
1	Programming Concepts and syntax	Fundamental logic
2	Cybersecurity Basics concepts	Intro security
3	Networking Fundamentals principles	Basic networking

4	Database Management principles	SQL and NoSQL
5	Cybersecurity Foundations cybersecurity concepts	Core
6	Cybersecurity Principles fundamentals	Security design
7	IT Systems Components operations	IT system roles/
8	Basic Scripting and Programming creation basics	Script/program
9	Basic Networking operate/analyze	Network build/
10	Network Defense protection	Network threat
11	Basic Cryptography usage	Cryptography
12	Operating Systems Concepts services	OS roles and
13	Advanced Network Technology & Protocols networking security	Advanced
14	Cloud Computing security	Cloud models and
15	Cybersecurity Ethics contexts	Ethics in cyber
16	Database Management Systems skills	DBMS utilization
17	Databases management, issues	DB usage,
18	Digital Communications comm protocols	Modern digital
19	Intrusion Detection/Prevention Systems detection/mitigation	Vulnerability
20	Low Level Programming programming	Secure low-level
21	Network Security Administration security infra	Enterprise
22	Network Technology and Protocols interactions	Common protocols/
23	Operating Systems Administration operations	OS system admin
24	Operating Systems Theory components/interfaces	OS concepts/
25	Vulnerability Analysis vulnerability coverage	System
26	Web Application Security practices	Web app tools and

Total KUs seeded: 26

COURSE ↔ KNOWLEDGE UNIT MAPPING (Many-to-Many)

Courses are mapped to multiple KUs to enable cross-institution matching.

Example mappings:

Course ID	Course Name	Mapped KU IDs
1	Data Structures & Algorithms Concepts)	KU 1 (Programming
2	Network Security Basics)	KU 2 (Cybersecurity
3	IT Fundamentals Fundamentals)	KU 3 (Networking
4	Web Development Management)	KU 4 (Database
23	Intro to Computer Security	KU 5, 6, 11, 15, 19, 25,
26		
13	Operating Systems	KU 7, 12, 14, 24
12	Computer Networking	KU 9, 13, 18, 22
17	Database Systems	KU 16, 17
24	Internet Security	KU 10, 21, 24, 26

Total course-KU links seeded: 32

→ This mapping is the core driver of the matching algorithm. Two courses from different institutions are considered equivalent when they share a sufficient overlap of Knowledge Units.

PROOF OF CONCEPT – TRANSFER REQUEST EXAMPLE

The following is a real end-to-end example seeded in the system:

STUDENT

Student ID : 1
Enrolled at: Roosevelt University (Institution 1)
Program : Computer Science (Program 1)

TRANSFER REQUEST

Request ID : 1
From : Roosevelt University (Institution 1)
To : Harold Washington College (Institution 4)
Course From : CS101 – Data Structures and Algorithms (Course 1, 4 credits)
Course To : SD102 – Web Development (Course 4, 3 credits)
Status : Pending
Date Submitted : 2026-02-16

COURSE MATCH RECORD

Match ID : 1
Institution From: Roosevelt University → Harold Washington College
Course From : CS101 – Data Structures and Algorithms
Course To : SD102 – Web Development
Match Status : Pending

→ When the system processes this request, it compares the KU profiles of CS101 and SD102. If enough KU overlap is found, the match is approved.

A director at Harold Washington College reviews and finalizes the decision.

PDF TRANSCRIPT PARSER

File: TranscriptPDF_reader/pdf_reader.py

PURPOSE:

Parses student academic transcripts in PDF format (Banner-based and web-saved PDFs) to extract:

- Course codes
- Course names
- Grades
- Credit hours

The extracted data is used to auto-populate transfer request submissions,
reducing manual entry errors and speeding up the matching workflow.

USAGE:

python pdf_reader.py --pdf <path_to_transcript.pdf>

SUPPORTED FORMATS:

- Banner-generated PDF transcripts
- Web-saved/printed PDF transcripts (with layout noise handling)

LIBRARIES USED:

- pdfplumber - for table-based extraction from structured PDFs
- re - for regex-based fallback parsing
- pandas - for structuring and exporting extracted data

DATA STANDARDIZATION AND VALIDATION LAYER

PURPOSE:

This layer supports the reliability of the Transfer Credit Match system by preparing, standardizing, and validating academic data before it is used by the KU matching process.

Since unofficial transcripts, course names, and KU labels may appear in different formats across institutions, this step helps ensure that the data entering the matching pipeline is clean, structured, and consistent.

WHY THIS LAYER IS NEEDED:

The parser may successfully extract course information from a transcript, but extracted data can still be incomplete, inconsistent, or differently phrased.

For example:

- course codes may be missing
- course titles may use different naming conventions
- KU-related terminology may vary across institutions
- some transcript rows may be partially extracted or noisy

Without standardization and validation, these inconsistencies can reduce the accuracy and trustworthiness of downstream KU matching.

CORE RESPONSIBILITIES:

- Standardize transcript-derived course data into a structured format
- Normalize KU terminology into a consistent internal naming scheme
- Validate parser output for missing or incomplete fields
- Generate synthetic/sample datasets for early-stage development
- Build benchmark cases for future evaluation of KU matching quality

EXAMPLE USE CASES:

1. KU Normalization
Different labels such as "binary arithmetic" and "base conversion" may refer to the same competency area. This layer maps such variants into a common KU label for consistency.
2. Parser Output Validation
Parsed transcript rows are checked for missing values such as course code, course title, credits, or malformed records before being used in matching.
3. Benchmark Dataset Creation
Sample and benchmark records are created so the team can test the matching pipeline even before large volumes of real data are available.

SAMPLE CODE SNIPPETS:

KU Normalization Example

```
-----  
ku_map = {  
    "binary arithmetic": "KU_BINARY_ARITH",  
    "base conversion": "KU_BINARY_ARITH",  
    "assembly language": "KU_ASSEMBLY",  
    "computer organization basics": "KU_COMP_ORG"  
}
```

```
raw_kus = ["binary arithmetic", "assembly language", "base  
conversion"]  
normalized_kus = [ku_map.get(ku.lower(), ku) for ku in raw_kus]
```

Parser Output Validation Example

```
-----  
import pandas as pd  
  
df = pd.DataFrame([  
    {"course_code": "CSC210", "course_title": "Assembly and Binary",  
"credits": 3},  
    {"course_code": None, "course_title": "Networking Basics",  
"credits": 3},  
    {"course_code": "CSC101", "course_title": None, "credits": 3}  
)  
  
df["missing_course_code"] = df["course_code"].isna()  
df["missing_course_title"] = df["course_title"].isna()  
df["missing_credits"] = df["credits"].isna()
```


VALUE TO THE SYSTEM:

This layer improves the overall quality of the Transfer Credit Match pipeline

by making the data cleaner, more consistent, and easier to evaluate. It also

gives the team a foundation for future analytics, confidence scoring, and KU matching performance assessment.

In short:

parser extracts the data,
this layer cleans and validates the data,
and the matching system uses better-quality input as a result.

DATA PIPELINE SUMMARY

[Student PDF Transcript]

↓

[pdf_reader.py] – parse courses from transcript

↓

[Transfer Request submitted to DB]

↓

[KU Mapping engine] – compare KU profiles of source vs target course

↓

[course_match record created] – status: Pending

↓

[Director reviews & approves/denies]

↓

[match_history updated] – audit trail preserved

CONTRIBUTORS (Data Science Branch)

- Data modeling and CSV seed data
- PDF transcript parsing pipeline
- Knowledge Unit taxonomy design
- Course-to-KU mapping for CAE-aligned programs

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END OF DOCUMENT

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